MIL-B-45451B (Ord)
18 August 1961
SUPERSEDING
MIL-B-45451A (Ord)
27 June 1960

MILITARY SPECIFICATION

BENITE STRANDS

1. SCOPE

1.1 This specification covers benite strands for use in ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-156 - Potassium Nitrate.
JAN-C-178 - Charcoal
JAN-N-244 - Nitrocellulose.
JAN-E-255 - Ethyl Centralite.
JAN-S-487 - Sulfur.
MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.
MIL-I-45208 - Inspection Requirements; General Specification for.
MIL-I-45607 - Inspection Equipment; Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-286 - Propellants; Standards for Methods of Sampling, Inspection and Testing.

FSC: 1375

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DRAWINGS

ORDNANCE CORPS

8830852 - Carton, Packing, for Benite Strands.
8830853 - Box Packing, for Benite Strands.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-8816619 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Picatinny Arsenal Technical report No. 2005 by Arnold O. Pallingston and Murray Weinstein, dated June 1954, relating to Methods of Calculation of Interior Ballistic Properties of Propellants from Closed Bomb Data. Picatinny Arsenal technical reports are available from Picatinny Arsenal, Dover, New Jersey.

Picatinny Arsenal Technical report No. 2367 by Arthur I. Rubin and August G. Edwards, dated December 1956, "The Use of Piezo Gages in Closed Bomb Tests of Solid Propellants." Picatinny Arsenal technical reports are available from Picatinny Arsenal, Dover, New Jersey.

3. REQUIREMENTS

3.1 Material.-The constituent materials used in the manufacture of benite strands shall comply with the following requirements.

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Constituent Material	Conforming to
Nitrocellulose (13.15 percent)	Specification JAN-N-244, Grade C, Type 1.
*Potassium Nitrate	Specification MIL-P-156, Class 1 or 2.
Sulfur	Specification JAN-S-487, Grade A.
Charcoal	Specification JAN-C-178, Class D.
Ethyl Centralite (added)	Specification JAN-E-255, Class 2.

*The potassium nitrate shall be pulverized to an average diameter of 20 plus or minus 10 microns as determined by the Fisher Sub-Sieve Sizer.

3.2 Composition.-The benite strands shall comply with the requirements specified in Table 1, when determined as specified in 4.3.1.

Table 1
Composition

Constituent	Percent
Nitrocellulose	40 plus or minus 1
Potassium Nitrate	44.3 plus or minus 1
Sulfur	6.3 plus or minus .3
Charcoal	9.4 plus or minus .3
Ethyl Centralite (added)	0.5 plus or minus .1

3.3 Total volatiles.-The total volatiles of the benite strands shall be not more than 2.25 percent maximum when determined as specified in 4.3.2.

3.4 Moisture content.-The moisture content of the benite strands shall be 1.00 percent, maximum, when tested as specified in 4.3.3.

3.5 Specific gravity.-The specific gravity of the benite strands shall be 1.65 plus or minus 0.15, when determined as specified in 4.3.4.

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3.6 Heat test (134.5 degrees Centigrade (C.).-The benite strands shall not change the color of normal methyl violet test paper to a salmon pink color in less than 35 minutes and shall not explode in less than 5 hours, when tested as specified in 4.3.5.

3.7 Straightness.-The benite strands shall be sufficiently straight to permit insertion without force, (finger pressure is permissible) into the gage. There shall be no chipping, scraping, cracking or similar damage to any of the strands when inserted into the gage upon testing as specified in 4.3.6.

3.8 Closed bomb quickness value.-The benite shall have a closed bomb quickness value (the time rate of pressure change (dp/dt) of 1,050 pounds per square inch (p.s.i.) per millisecond, maximum, when tested as specified in 4.3.7.

3.9 Foreign matter.-The benite strands shall be free of foreign matter such as wood, paper, metal particles, dirt, oil, grease, etc.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-G-2550 shall apply.

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4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

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4.2 Inspection provisions.

4.2.1 Lot formation.-A lot consists of one or more batches of benite strands of one composition designation only, produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of benite strands that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. In addition, each lot shall contain:

a. Potassium nitrate of one type only, from one lot.
b. Charcoal of one type only, from one lot.
c. Nitrocellulose of one type only, from one lot.
d. Ethyl centralite of one type only, from one lot.
e. Sulfur of one type only, from one lot.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Benite strands (see applicable drawing (dwg.).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	None defined	
Minor:	AQL 1.50 percent	
*201.	Straightness (see 4.3.6)	Gage
202.	Length	Gage
203.	Diameter	Gage

4.2.2.2 Unsealed carton (see dwg. 8830852).

Categories	Defects	Method of inspection
Critical:	None defined.	

*Note: A charge shall be the unit of inspection.

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Major:	AQL 0.25 percent	
101.	Filler or cushion missing	Visual
102.	Foreign matter (see 3.8).....	Visual
Minor:	AQL 0.40 percent	
201.	Carton not full	Visual

4.2.2.3 Sealed carton (see dwg. 8830852).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Sealing strip missing or insecure	Visual-Manual
102.	Carton punctured or torn	Visual
Minor:	AQL 0.40 percent	
201.	Marking misleading or unidentifiable ...	Visual

4.2.2.4 Box, packing (see dwg. 8830853).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Contents exposed	Visual
102.	Contents loose	Manual
Minor:	AQL 1.50 percent	
201.	Strapping missing, broken or loose	Visual-Manual
202.	Board broken or split	Visual
203.	Marking misleading or unidentifiable ...	Visual

4.2.3 Testing.

4.2.3.1 Chemical tests (see 3.1 through 3.6 and 6.3).-Ten pounds of the benite strands shall be selected at random from each lot. Five pounds shall be used for the chemical and stability tests to be conducted at a Government approved laboratory in accordance with the procedures specified in 4.3 of this specification. The remaining five pounds shall be forwarded to the

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Commanding Officer, Picatinny Arsenal, Dover, New Jersey, ATTN: ORDBB-TM7, so that the 65.5 degrees C. surveillance test may be conducted for informational purposes. The surveillance test sample shall be packed for shipment as specified in 6.3. If the sample fails to comply with any of the requirements specified, the lot shall be rejected.

4.2.3.2 Closed bomb test (see 3.8 and 4.3.7) Ballistic sample.-The sample of benite shall consist of six (6) sample units, each of sufficient amount to permit testing in triplicate. The sample units shall be randomly drawn from the entire lot in a manner such that every portion of the lot has an equal chance of selection. The mean of triplicate test results for each sample unit shall be considered as an individual quickness value. If any individual quickness value exceeds the applicable lot requirement by more than 5 percent, the lot shall be rejected.

4.2.4 Inspection equipment.-For the performance of the inspections listed herein, the contractor shall have available and utilize correctly all necessary gages, measuring and testing equipment identified by Index of Inspection Equipment Lists No. IEL-8816619, or approved equivalents thereto. Designs of equipment developed by the contractor for use in lieu of that specified in the Index of Inspection Equipment Lists shall be documentated and submitted to the Government for approval as equivalent (see 6.2). Approval shall be obtained prior to initiation of production. Supply and maintenance of inspection equipment shall be in accordance with Specification MIL-L-45607.

4.3 Test methods and procedures.

4.3.1 Determination of composition.

4.3.1.1 Nitrocellulose.-The percentage of nitrocellulose shall be determined in accordance with method No. 209.2. of Standard MIL-STD-286.

4.3.1.2 Potassium nitrate.-The percentage of potassium nitrate shall be determined in accordance with Method No. 310.4 of Standard MIL-STD-286.

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4.3.1.3 Charcoal

4.3.1.3.1 Standard method for charcoal.-The percentage of charcoal shall be determined as follows: Accurately weigh a 1 gram (gm.) sample into 250 milliliter (ml.) beaker, add 75 ml. of concentrated nitric acid and digest for two hours in accordance with Method No. 309.1 of Standard MIL-STD-286. Filter through a tared gooch filtering crucible and wash the residue in the crucible with hot acetic acid until the addition of an equal volume of water to the filtrate fails to give a precipitate. Dry the crucible and contents at 100 degrees C. for two hours, cool in a desiccator and weigh. Calculate the increase in weight as follows:

$$\text{Charcoal} = \frac{(A-B) \times 100}{W}$$

where:

A = final weight of crucible.
B = initial weight of crucible.
W = weight of sample.

4.3.1.3.2 Alternate method for charcoal.-Accurately weigh a 1.00 gm. plus or minus 10 milligrams into a 250 ml. beaker, add 5 ml. of petroleum ether and 30 ml. of pyridine. Heat on a steam bath with occasional stirring until the soluble ingredients are dissolved. Remove from heat, add 75 ml. of acetone, mix well, cover with a watch glass, and let stand for one hour. Decant into a Silas crucible No. 3010 or equal and filter with suction. Add hot glacial acetic acid to the crucible if at any time the filtration becomes too slow. Transfer the solids from the beaker to the crucible by means of acetone from a wash bottle. Wash the crucible and contents well with acetone. Wash with approximately 100 ml. of hot water allowing one or more portions of the water to stand in the crucible for one to two minutes. Dry for one hour at 100 degrees C., cool, and weigh. Calculate the charcoal as follows:

$$\text{Percent charcoal} = \frac{\text{gain in weight of crucible} \times 100}{\text{weight of sample}}$$

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4.3.1.4 Ethyl centralite.-The percentage of ethyl centralite shall be as determined in accordance with Method No. 217.2 of Standard MIL-STD-286.

4.3.1.5 Sulfur.-The percentage of sulfur shall be determined as follows: A sample of 0.6 to 0.9 gms. containing approximately 0.075 gm. of sulfur shall be accurately weighed into a 250 ml. beaker and digested with 10 ml. of nitric acid for 24 hours on a steam bath, adding more acid if needed to maintain the volume. One to two ml. of bromine shall then be added and the beaker heated on a steam bath until the excess bromine is entirely evaporated. If any undissolved matter remains, the bromination shall be repeated. When all bromine is eliminated, the acid residue shall be diluted to approximately 150 ml. with water heated to boiling, and 10 ml. of 10 percent barium chloride solution added dropwise with stirring. The precipitated solution shall be heated for at least one hour on a steam bath, filtered through a tared gooch crucible, washed with water, dried and ignited at 600 degrees C. for one-half hour. The percentage of sulfur shall be calculated as follows:

$$\text{Sulfur, percent} = \frac{13.74A}{W}$$

where:
A = weight of precipitate (barium sulfate).
W = weight of sample.

4.3.2 Determination of total volatiles.-The percentage of total volatiles shall be in accordance with Method No. 103.3 of Standard MIL-STD-286.

4.3.3 Determination of total moisture content.-The determination of total moisture content shall be in accordance with Method No. 102.1 of Standard MIL-STD-286.

4.3.4 Determination of specific gravity.-The determination for specific gravity shall be in accordance with Method No. 510.1 of Standard MIL-STD-286.

4.3.5 Heat test (134.5 degrees C.).-The 134.5 degrees C. heat test shall be performed as specified in Method No. 404.1 of Standard MIL-STD-286.

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4.3.6 Straightness (see 4.2.2.1).-The applicable number of benite strands shall enter and/or leave the gage without having to be forced. It is permissible to use finger pressure or to rotate a strand to permit the charge to enter provided no damage to any strand occurs.

4.3.7 Closed bomb test.-The benite strands shall be cut into segments one half inch long plus or minus 1/8 inch and subjected to a conditioning temperature of 70 degrees Fahrenheit (F.) plus or minus 2 degrees F. for a period of four hours. The testing shall be performed utilizing a 200 ml. plus or minus 10 ml. closed bomb assembled with a piezo-electric gauge, a cathode ray oscilloscope, a recording camera and associated electrical apparatus for making a permanent firing record. Each individual quickness value shall be determined at 5000 p.s.i. bomb pressure when fired in a closed bomb loaded to a density of 0.1 gm/c.c and be expressed in terms of p.s.i./ milliseconds. The quickness value shall be calculated as follows:

$$dP/dt = K \frac{Vy}{Ry}$$

where:
V_y = measured voltage in the rate circuit.
R_y = resistance of the rate circuit.
K = the gage constants in p.s.i./coulomb inches square.

The determination of pressure can be calculated at the 5000 p.s.i. point as follows:

$$P = K C_x V_x$$

where:
C_x = the capacitance of the pressure circuit.
V_x = the voltage of the pressure circuit.

Note: Calibration of the Piezo Gage shall be performed in an oil medium by quick release of the dead weight from a piston of known surface area (see Picatinny Arsenal Technical Report 2367).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C.-The benite strands shall be packed in accordance with dwg. C8830852.

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5.2 Packing.

5.2.1 Level C.-The benite strands shall be packed in accordance with dwg. C8830853.

5.2.2 Marking.-Marking shall be in accordance with dwg. F8796522.

5.3 Data cards.-Data card information shall be as specified in Specification MIL-Q-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Submission of contractor equipment designs.-Documentation of contractor developed equipment designs shall be submitted in duplicate to Picatinny Arsenal, Dover, New Jersey, ATTN: ORDBB-ND, for approval. This documentation shall identify all standard devices and set-ups etc., as well as special equipment designs.

6.3 The surveillance testing shall be performed by a Government installation. All samples shall be packed in airtight containers and shall be marked to show the propellant designation, lot number, manufacturer, date of sampling, contract number and number of pounds in the lot.

6.4 Closed bomb data.-Picatinny Arsenal Technical Report No. 2005.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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